

Тема: Настройка протокола STP (IEEE 802.1D)

1) Для заданной на схеме schema-lab2 сети, состоящей из управляемых коммутаторов и персональных компьютеров настроить протокол STP, назначив явно один из коммутаторов корневым настройкой приоритета

```
vIOS-L2-01#enable
vIOS-L2-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
vIOS-L2-01(config)#spanning-tree vlan 1 priority 4096
vIOS-L2-01(config)#exit
```

2) Проверить доступность каждого с каждым всех персональных компьютеров (VPCS), результаты запротоколировать

Назначение ip:

```
PC1> ip 192.168.1.1/24
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0
```

```
PC2> ip 192.168.1.2/24
Checking for duplicate address...
PC2 : 192.168.1.2 255.255.255.0
```

```
PC3> ip 192.168.1.3/24
Checking for duplicate address...
PC3 : 192.168.1.3 255.255.255.0
```

```
PC4> ip 192.168.1.4/24
Checking for duplicate address...
PC4 : 192.168.1.4 255.255.255.0
```

```
PC5> ip 192.168.1.5/24
Checking for duplicate address...
PC5 : 192.168.1.5 255.255.255.0
```

```
PC6> ip 192.168.1.6/24
Checking for duplicate address...
PC6 : 192.168.1.6 255.255.255.0
```

Проверка доступности:
PC1:

```
PC1> ping 192.168.1.2
```

```
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=8.317 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=5.645 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=9.932 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=18.123 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=13.028 ms
```

PC1> ping 192.168.1.3

```
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=25.685 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=28.369 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=33.356 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=41.648 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=40.669 ms
```

PC1> ping 192.168.1.4

```
84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=27.158 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=25.768 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=41.237 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=23.429 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=84.987 ms
```

PC1> ping 192.168.1.5

```
84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=25.924 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=19.784 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=12.341 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=34.047 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=30.843 ms
```

PC1> ping 192.168.1.6

```
84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=20.630 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=52.186 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=28.271 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=29.598 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=31.873 ms
```

PC2:

PC2> ping 192.168.1.1

```
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=6.992 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=13.633 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=9.957 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=9.342 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=13.922 ms
```

PC2> ping 192.168.1.3

```
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=34.380 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=27.713 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=32.069 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=28.228 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=40.221 ms
```

PC2> ping 192.168.1.4

```
84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=46.258 ms
```

```
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=27.078 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=61.409 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=18.410 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=45.813 ms
```

```
PC2> ping 192.168.1.5
```

```
84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=25.031 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=9.930 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=50.515 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=33.625 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=14.586 ms
```

```
PC2> ping 192.168.1.6
```

```
84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=35.039 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=11.417 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=13.161 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=44.950 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=30.393 ms
```

```
PC3:
```

```
PC3>
```

```
PC3> ping 192.168.1.1
```

```
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=31.599 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=21.400 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=31.068 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=18.017 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=22.455 ms
```

```
PC3> ping 192.168.1.2
```

```
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=50.317 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=33.889 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=32.179 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=23.259 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=28.125 ms
```

```
PC3> ping 192.168.1.4
```

```
84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=14.079 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=6.941 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=5.123 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=19.670 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=6.960 ms
```

```
PC3> ping 192.168.1.5
```

```
84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=23.954 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=18.485 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=29.024 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=33.243 ms
```

84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=14.273 ms

PC3> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=18.094 ms

84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=21.611 ms

84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=30.755 ms

84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=38.527 ms

84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=25.166 ms

PC4:

PC4> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=23.503 ms

84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=34.440 ms

84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=35.781 ms

84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=26.346 ms

84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=19.647 ms

PC4> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=28.800 ms

84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=27.803 ms

84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=42.882 ms

84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=16.678 ms

84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=49.314 ms

PC4> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=13.011 ms

84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=9.782 ms

84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=5.496 ms

84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=16.531 ms

84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=14.843 ms

PC4> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=28.205 ms

84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=36.088 ms

84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=25.890 ms

84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=43.011 ms

84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=39.140 ms

PC4> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=26.554 ms

84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=13.586 ms

84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=27.483 ms

84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=28.017 ms

84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=29.661 ms

PC5:

PC5> ping 192.168.1.1

```
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=18.117 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=15.156 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=39.572 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=27.028 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=25.487 ms
```

PC5> ping 192.168.1.2

```
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=26.483 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=35.191 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=16.714 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=24.436 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=26.983 ms
```

PC5> ping 192.168.1.3

```
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=26.694 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=13.608 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=25.692 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=37.624 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=12.006 ms
```

PC5> ping 192.168.1.4

```
84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=32.516 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=19.653 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=29.347 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=30.949 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=22.437 ms
```

PC5> ping 192.168.1.6

```
84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=14.237 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=4.255 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=16.562 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=10.294 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=11.618 ms
```

PC6:

PC6> ping 192.168.1.1

```
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=11.128 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=31.394 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=33.732 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=40.974 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=27.092 ms
```

PC6> ping 192.168.1.2

```
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=15.582 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=31.855 ms
```

```
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=43.404 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=29.512 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=39.142 ms
```

```
PC6> ping 192.168.1.3
```

```
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=31.015 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=41.679 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=23.779 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=27.653 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=25.554 ms
```

```
PC6> ping 192.168.1.4
```

```
84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=43.404 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=22.522 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=40.481 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=31.443 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=21.783 ms
```

```
PC6> ping 192.168.1.5
```

```
84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=8.807 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=9.472 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=10.218 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=1.924 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=6.745 ms
```

3) На изображении схемы отметить BID каждого коммутатора и режимы работы портов (RP/DP/blocked) и стоимости маршрутов, результат сохранить в файл

Layer1-Switch-1:

```
vIOS-L2-01>show spanning-tree
```

VLAN0001

Spanning tree enabled protocol ieee

```
Root ID      Priority    4097
             Address    0cb1.51cd.0000
             This bridge is the root
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
```

```
Bridge ID   Priority    4097 (priority 4096 sys-id-ext 1)
             Address    0cb1.51cd.0000
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----	----	---	-----	-----	
Gi0/0	Desg	FWD	4	128.1	Shr
Gi0/1	Desg	FWD	4	128.2	Shr
Gi0/2	Desg	FWD	4	128.3	Shr
Gi0/3	Desg	FWD	4	128.4	Shr

Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr
Gi1/2	Desg	FWD	4	128.7	Shr
Gi1/3	Desg	FWD	4	128.8	Shr

Layer1-Switch-2:

vIOS-L2-01>show spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID	Priority	4097			
	Address	0cb1.51cd.0000			
	Cost	4			
	Port	1 (GigabitEthernet0/0)			
	Hello Time	2 sec	Max Age 20 sec	Forward Delay 15 sec	

Bridge ID	Priority	32769	(priority 32768 sys-id-ext 1)		
	Address	0c29.4fae.0000			
	Hello Time	2 sec	Max Age 20 sec	Forward Delay 15 sec	
	Aging Time	300 sec			

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----	-----	---	-----	-----	
-----	-----	---	-----	-----	
Gi0/0	Root	FWD	4	128.1	Shr
Gi0/1	Altn	BLK	4	128.2	Shr
Gi0/2	Desg	FWD	4	128.3	Shr
Gi0/3	Desg	FWD	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr
Gi1/2	Altn	BLK	4	128.7	Shr
Gi1/3	Altn	BLK	4	128.8	Shr

Layer2-Switch-3:

vIOS-L2-01>show spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID	Priority	4097			
	Address	0cb1.51cd.0000			
	Cost	4			
	Port	1 (GigabitEthernet0/0)			
	Hello Time	2 sec	Max Age 20 sec	Forward Delay 15 sec	

Bridge ID	Priority	32769	(priority 32768 sys-id-ext 1)		
	Address	0cdd.ed94.0000			
	Hello Time	2 sec	Max Age 20 sec	Forward Delay 15 sec	
	Aging Time	300 sec			

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----	-----	---	-----	-----	
-----	-----	---	-----	-----	
Gi0/0	Root	FWD	4	128.1	Shr
Gi0/1	Altn	BLK	4	128.2	Shr

Gi0/2	Altn	BLK	4	128.3	Shr
Gi0/3	Altn	BLK	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr

Layer2-Switch-4:

vIOS-L2-01>show spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID	Priority	4097			
	Address	0cb1.51cd.0000			
	Cost	4			
	Port	1 (GigabitEthernet0/0)			
	Hello Time	2 sec	Max Age 20 sec	Forward Delay 15 sec	

Bridge ID	Priority	32769	(priority 32768 sys-id-ext 1)		
	Address	0cf8.35a7.0000			
	Hello Time	2 sec	Max Age 20 sec	Forward Delay 15 sec	
	Aging Time	300 sec			

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----	----	---	-----	-----	
-----	----	---	-----	-----	
Gi0/0	Root	FWD	4	128.1	Shr
Gi0/1	Altn	BLK	4	128.2	Shr
Gi0/2	Altn	BLK	4	128.3	Shr
Gi0/3	Altn	BLK	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr

Layer2-Switch-5:

vIOS-L2-01>show spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID	Priority	4097			
	Address	0cb1.51cd.0000			
	Cost	4			
	Port	1 (GigabitEthernet0/0)			
	Hello Time	2 sec	Max Age 20 sec	Forward Delay 15 sec	

Bridge ID	Priority	32769	(priority 32768 sys-id-ext 1)		
	Address	0c05.5640.0000			
	Hello Time	2 sec	Max Age 20 sec	Forward Delay 15 sec	
	Aging Time	300 sec			

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----	----	---	-----	-----	
-----	----	---	-----	-----	
Gi0/0	Root	FWD	4	128.1	Shr
Gi0/1	Altn	BLK	4	128.2	Shr
Gi0/2	Desg	FWD	4	128.3	Shr

Gi0/3	Desg FWD 4	128.4	Shr
Gi1/0	Desg FWD 4	128.5	Shr
Gi1/1	Desg FWD 4	128.6	Shr

4) При помощи Wireshark отследить передачу пакетов hello от корневого коммутатора на всех линках (nb!), результаты включить в отчет

Пакет hello, перехваченный на линке от Корневого коммутатора (возраст 0)
 Frame 934: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:b1:51:cd:00:00

Root Path Cost: 0

Bridge Identifier: 4096 / 1 / 0c:b1:51:cd:00:00

Port identifier: 0x8003

Message Age: 0

Max Age: 20

Hello Time: 2

Forward Delay: 15

Пакет hello, перехваченный на заблокированном линке между Switch-2 и Switch-3 (возраст 1)

Frame 1025: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:b1:51:cd:00:00

Root Path Cost: 4

Bridge Identifier: 32768 / 1 / 0c:29:4f:ae:00:00

Port identifier: 0x8003

Message Age: 1

Max Age: 20

Hello Time: 2

Forward Delay: 15

Пакеты hello присутствуют на всех линках, включая заблокированные

5) Изменить стоимость маршрута для порта RP произвольного назначенного (designated) коммутатора, повторить действия из п.3, результат сохранить в отдельный файл

Layer2-Switch-3:

vIOS-L2-01#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

```

vIOS-L2-01(config)#interface Gi0/0
vIOS-L2-01(config-if)#spanning-tree vlan 1 cost 100
vIOS-L2-01(config-if)#end

```

```

vIOS-L2-01#show spanning-tree

```

```

VLAN0001

```

```

Spanning tree enabled protocol ieee

```

```

Root ID      Priority      4097
             Address      0cb1.51cd.0000
             Cost        4
             Port        2 (GigabitEthernet0/1)
             Hello Time   2 sec    Max Age 20 sec    Forward Delay 15 sec

```

```

Bridge ID    Priority      32769 (priority 32768 sys-id-ext 1)
             Address      0cdd.ed94.0000
             Hello Time   2 sec    Max Age 20 sec    Forward Delay 15 sec
             Aging Time   300 sec

```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Altn	BLK	100	128.1	Shr
Gi0/1	Root	FWD	4	128.2	Shr
Gi0/2	Altn	BLK	4	128.3	Shr
Gi0/3	Altn	BLK	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr

Порт Gi0/0 стал не выгоден из-за высокой стоимости, поэтому теперь он заблокирован, а Gi0/1 стал Root.

6) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств

```

Layer1-Switch-1:

```

```

vIOS-L2-01>enable

```

```

vIOS-L2-01#write memory

```

```

Building configuration...

```

```

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z
86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU
hog, PC 0x00889D77

```

```

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z
86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU
hog, PC 0x00889D77

```

```

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z
86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU
hog, PC 0x00889D77

```

```

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z
86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU

```

hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77

Compressed configuration from 5276 bytes to 2029 bytes

*Sep 1 14:38:04.676: %SYS-3-CPUHOG: Task is running for (2000)msecs, more than (2000)msecs (37/12),process = Async write process.

*Sep 1 14:38:07.284: %SYS-3-CPUHOG: Task is running for (1999)msecs, more than (2000)msecs (29/0),process = Async write process.

*Sep 1 14:38:12.000: %SYS-3-CPUHOG: Task is running for (1997)msecs, more than (2000)msecs (43/22),process = Async write process.

*Sep 1 14:38:14.708: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than (2000)msecs (76/45),process = Async write process.

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 86CE9Ez 8679B9z 8574D3z 85914Ez - Process "Exec", CPU hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 889E8Dz 889D8Dz 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889FE2

[OK]

*Sep 1 14:38:17.348: %SYS-3-CPUHOG: Task is running for (1997)msecs, more than (2000)msecs (32/0),process = Async write process.

*Sep 1 14:38:19.752: %SYS-3-CPUHOG: Task is running for (1997)msecs, more than (2000)msecs (70/36),process = Exec.

*Sep 1 14:38:22.560: %SYS-3-CPUHOG: Task is running for (1997)msecs, more than (2000)msecs (32/0),process = Async write process.

*Sep 1 14:38:22.706: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...

*Sep 1 14:38:24.335: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.

Layer1-Switch-2:

vIOS-L2-01>enable

vIOS-L2-01#write memory

Building configuration...

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 889E8Dz 889D8Dz 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889FE2

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z
86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU
hog, PC 0x00889D77

Compressed configuration from 5239 bytes to 2001 bytes

*Sep 1 14:37:46.297: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than
(2000)msecs (37/3),process = Async write process.

*Sep 1 14:37:48.605: %SYS-3-CPUHOG: Task is running for (1997)msecs, more than
(2000)msecs (68/38),process = Async write process.

*Sep 1 14:37:50.885: %SYS-3-CPUHOG: Task is running for (1997)msecs, more than
(2000)msecs (26/0),process = Async write process.

*Sep 1 14:37:54.461: %SYS-3-CPUHOG: Task is running for (1999)msecs, more than
(2000)msecs (71/50),process = Async write process.

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z
86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU
hog, PC 0x00889D77

[OK]

*Sep 1 14:37:57.489: %SYS-3-CPUHOG: Task is running for (1997)msecs, more than
(2000)msecs (35/0),process = Async write process.

*Sep 1 14:38:01.865: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than
(2000)msecs (27/0),process = Async write process.

*Sep 1 14:38:02.427: %GRUB-5-CONFIG_WRITING: GRUB configuration is being
updated on disk. Please wait...

vIOS-L2-01#

*Sep 1 14:38:04.437: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to
disk successfully.

Layer2-Switch-3:

vIOS-L2-01#write memory

Building configuration...

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z
867

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z
860

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z
867

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z
867

Compressed configuration from 5072 bytes to 1946 bytes[OK]

*Sep 1 15:57:23.874: %SYS-3-CPUHOG: Task is running for (2000)msecs, more
than.

*Sep 1 15:57:26.558: %SYS-3-CPUHOG: Task is running for (1997)msecs, more
than.

*Sep 1 15:57:29.362: %SYS-3-CPUHOG: Task is running for (1999)msecs, more
than.

*Sep 1 15:57:31.810: %SYS-3-CPUHOG: Task is running for (2000)msecs, more
than.

*Sep 1 15:57:35.701: %GRUB-5-CONFIG_WRITING: GRUB configuration is being
updat.

vIOS-L2-01#

*Sep 1 15:57:37.052: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to.

Layer2-Switch-4:
vIOS-L2-01>enable
vIOS-L2-01#write memory
Building configuration...

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 867

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 867

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 867

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 867

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 860

Compressed configuration from 5041 bytes to 1930 bytes[OK]

*Sep 1 15:56:00.250: %SYS-3-CPUHOG: Task is running for (1997)msecs, more than.

*Sep 1 15:56:02.486: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than.

*Sep 1 15:56:06.574: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than.

*Sep 1 15:56:09.686: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than.

*Sep 1 15:56:12.470: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than.

*Sep 1 15:56:16.544: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updat.

*Sep 1 15:56:18.249: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to.

Layer2-Switch-5:
vIOS-L2-01>enable
vIOS-L2-01#write memory
Building configuration...

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x008B5430

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x008B5430

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x008B5430

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77

Compressed configuration from 5041 bytes to 1927 bytes

*Sep 1 15:55:45.267: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than (2000)msecs (17/3),process = Async write process.

*Sep 1 15:55:48.387: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than (2000)msecs (16/0),process = Async write process.

*Sep 1 15:55:50.971: %SYS-3-CPUHOG: Task is running for (2000)msecs, more than (2000)msecs (42/20),process = Async write process.

*Sep 1 15:55:53.515: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than (2000)msecs (18/0),process = Async write process.

*Sep 1 15:55:55.807: %SYS-3-CPUHOG: Task is running for (1999)msecs, more than (2000)msecs (34/20),process = Async write process.

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x008B5430

[OK]

*Sep 1 15:55:58.567: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than (2000)msecs (15/0),process = Async write process.

*Sep 1 15:56:02.251: %SYS-3-CPUHOG: Task is running for (1999)msecs, more than (2000)msecs (57/37),process = Async write process.

*Sep 1 15:56:02.909: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...

*Sep 1 15:56:04.483: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.

PC1> save

Saving startup configuration to startup.vpc
. done

PC2> save

Saving startup configuration to startup.vpc
. done

PC3>

PC3> save

Saving startup configuration to startup.vpc
. done

PC4> save

Saving startup configuration to startup.vpc
. done

PC5> save

```
Saving startup configuration to startup.vpc  
. done
```

```
PC6> save  
Saving startup configuration to startup.vpc  
. done
```